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Hybrid Dipping–Shower Wet Scrubber for Multi-Pollutant Emission Reduction From Marine Diesel Exhaust: Experimental Assessment of a Low-Cost Prototype as an Auxiliary Exhaust Gas Cleaning Approach

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Abstract: Marine diesel engines are major contributors to atmospheric pollution, emitting SO_2, CO_2, NO_x, and CO in quantities subject to increasingly stringent International Maritime Organization (IMO) regulations. Wet scrubbers are a cost-effective exhaust gas cleaning option, yet most existing systems employ either immersion (dipping) or spray (shower) gas–liquid contact in isolation, and experimental data on a structurally combined dipping–shower contact pattern under marine diesel conditions remain scarce. In this study, the term "hybrid" denotes this structural combination of bubble-immersion and spray contact, not the open-loop/closed-loop switching systems commonly called hybrid scrubbers in shipping. A prototype hybrid wet scrubber was fabricated from recycled industrial oil-storage drums and tested on a Dongfeng diesel engine at four engine-speed settings (25%, 50%, 75%, and 100% of rated speed). Concentrations of CO_2, SO_2, NO_2, and CO were measured at the scrubber inlet and outlet with an Internet of Things (IoT)-based multi-gas sensor node under three configurations: dipping-only, shower-only, and hybrid (dipping + shower), with five consecutive sensor readings per condition, and removal efficiency (η) was reported as mean ± standard deviation. Averaged over the four engine speeds, the hybrid configuration gave the highest removal of all four gases, with mean gains of about 6 and 10 percentage points for SO_2 and about 3 and 10 percentage points for CO relative to dipping-only and shower-only operation, respectively; it was not superior at every individual condition (e.g., CO at 100% and CO_2 at 75% of rated speed). SO_2 achieved the highest removal (75–79% in hybrid mode, coefficient of variation below 7%), and CO peaked at 78% at 25% rated speed, declining at higher speeds because of reduced gas–liquid contact time. By contrast, CO_2 removal remained inherently limited (8–11%) owing to its low aqueous solubility, and NO_2 removal was below 10%, reflecting the poor water solubility of NO that dominates NO_x. The overall efficiency trend was SO_2 > CO > CO_2 > NO_2. These results indicate that the hybrid dipping–shower scrubber is best regarded as a low-cost, transitional emission-control device suited mainly to SO_2 and partial CO abatement; it cannot meet deep CO_2 capture or stringent NO_x-control requirements without alkaline absorbents, oxidants, or multi-stage treatment. Future work should address chemically enhanced scrubbing liquids, repeated long-duration testing, and characterization of the wash water to manage the risk of transferring pollution from air to water.

Keywords: Hybrid wet scrubber; Bubble-spray wet scrubber; Combined gas–liquid contact; Marine diesel emission; Exhaust gas cleaning system; SO_2 removal; Multi-pollutant abatement

1 Introduction

The International Maritime Organization (IMO) adopted its Initial Strategy on the reduction of greenhouse gas (GHG) emissions from ships in April 2018 (Resolution MEPC.304(72)), which aimed to reduce the carbon intensity of international shipping by at least 40% by 2030 and its total annual GHG emissions by at least 50% by 2050, both relative to 2008 [1]. In July 2023, the Initial Strategy was replaced by the 2023 IMO Strategy (Resolution MEPC.377(80)), which raised the ambition to net-zero GHG emissions by or around 2050 and introduced indicative checkpoints of at least a 20% reduction (striving for 30%) by 2030 and at least a 70% reduction (striving for 80%)

by 2040, relative to 2008 [2, 3]. Modelling of the 2023 targets indicates that they are broadly consistent with a well-below 2 °C trajectory but fall short of what is required to limit warming to 1.5 °C [4]. Because the eventual contribution of shipping to global warming depends on the cumulative emissions released until the sector reaches net zero, the pace of near-term reductions matters as much as the 2050 end point [5]. Figure 1 presents cumulative well-to-wake GHG emissions of international shipping from 2020 to 2050 for the baseline, IMO Minimum, IMO Striving and 1.5 °C pathways. The bars separate unavoided emissions and avoided emissions from operational efficiency and GFI standards, while dashed lines denote the 2 °C, 1.7 °C, and 1.5 °C carbon budgets [4].

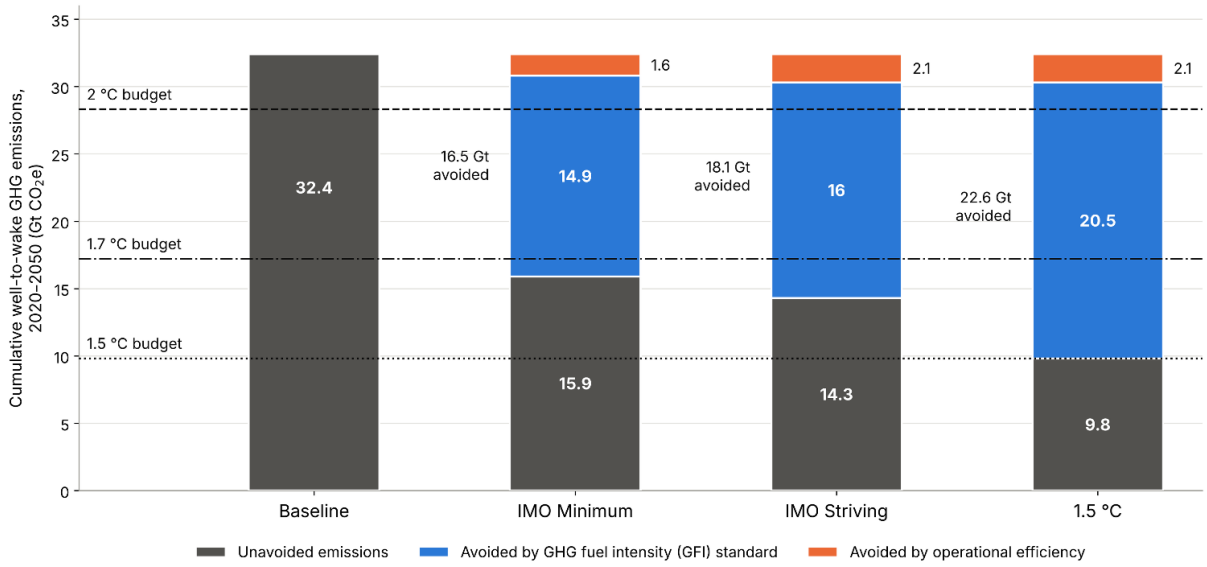


Figure 1. Cumulative well-to-wake GHG emissions from international shipping (2020–2050) under baseline, IMO Minimum, IMO Striving, and 1.5 °C policy scenarios

Note: IMO = International Maritime Organization; GHG = greenhouse gas; GFI = GHG Fuel Intensity.

Strategies and preventive measures to reduce GHG emissions have been implemented by the IMO in short, medium, and longterm programs. These measures are being taken to achieve the 2030 and 2040 targets [3]. The exhaust of a ship's main engine contains both GHGes and air pollutants. Carbon dioxide (CO_2) is the principal GHG, whereas sulfur dioxide (SO_2) and nitrogen oxides (NO_x = NO + NO_2) are air pollutants rather than direct GHGes: SO_2 forms sulfate aerosols that cause acidification and exert a net cooling rather than warming effect, and NO_x contributes to ozone and secondary particle formation and to acid deposition. NO_x should not be confused with nitrous oxide (N_2O), which is a GHG [6].

Effective emission control technology is necessary to lessen the detrimental effects of the shipping industry's substantial contribution to environmental pollution through exhaust emissions from the burning of fossil fuels in ship engines [7]. One of the Exhaust Gas Cleaning Systems technologies that is frequently used to lower SO_2 emissions to the 0.5% threshold in compliance with international regulations is wet scrubbers [8, 9]. In order to capture SO_x and solid particulates prior to their release into the atmosphere, the Wet Scrubber system directs exhaust gases through high-pressure seawater or freshwater spray [10]. The alkalinity of seawater or other chemicals in the freshwater system neutralize the sulfuric acid that is created chemically from a SO_2 solution in the wet scrubber process [11, 12]. To guarantee adherence to emission regulations, this system typically consists of residue handling facilities, wash water treatment installations, scrubber units positioned around the stack, and continuous emission monitoring systems [13]. There are three primary categories of wet scrubber systems: open loop, which uses seawater as a scrubbing medium; closed loop, which uses fresh water mixed with sodium hydroxide; and hybrid systems, which combine the two [14, 15]. This hybrid system offers high operational flexibility as it allows switching between open loop and closed loop modes depending on water conditions or local emission regulations [16, 17].

Generally, standard wet scrubbers have a negligible effect on CO_2 removal from ship exhaust [18]. This is primarily due to the low solubility of CO_2 in the aqueous solutions typically used in these scrubbers, such as seawater [18]. Wet scrubbing processes utilizing certain urea solutions have shown less than 10% CO_2 absorption efficiency [19].

Wet scrubbers demonstrate high efficiency in removing SO_x from ship exhaust. Removal rates often exceed 95%, with some studies reporting up to 98% reduction [20]. Freshwater scrubbers, for instance, are capable of reducing SO_x by up to 97% [12, 15]. To achieve emission levels comparable to using 0.1% sulfur fuel with a 3.5% sulfur

fuel, scrubbers need to remove approximately 97.1% of SO_x [21]. Electrified scrubbers can further enhance the absorption efficiency of SO_2 [18].

In general, nitrogen oxides, such as nitric oxide and nitrogen dioxide, cannot be effectively reduced by wet scrubbers [22]. The primary issue is that nitric oxide, which makes up a significant amount of NO_x emissions (approximately 90%), is difficult to scrub and has very low water solubility [23]. Technologies like Selective Catalytic Reduction are commonly used to significantly reduce NO_x in marine exhaust. To effectively address both SO_x and NO_x emissions, these systems are frequently coupled with scrubbers [24].

Wet scrubbers are also utilized to lower exhaust outlet operating temperatures. The purpose of wet scrubber systems is to drastically lower the temperature of gas emissions. For instance, in the primary quenching stage of a multi-stage scrubbing process, gas temperatures can decrease from roughly 900°C to 450°C [25]. This temperature drop is relevant for two distinct reasons. First, when the heat removed from the exhaust is recovered upstream of the scrubber, for example in an exhaust-gas economizer or heat-recovery unit, it can raise overall boiler or plant efficiency [26]; simply quenching the gas inside a scrubber, as in the present prototype, removes this heat without recovering it. Second, a lower gas temperature increases the solubility of SO_2 in the scrubbing liquid and therefore favours its absorption [27].

There is a large research gap in the thorough evaluation and optimization of sophisticated wet scrubber technologies, such as the shower and dip wet scrubber, for the efficient removal of various contaminants from marine diesel exhaust under dynamic marine operating conditions. This gap underlines the significance of resolving crucial elements such as hydrodynamics, mass transfer, solvent stability, long-term performance, and realistic industrial scale deployment.

Current literature underscores a substantial necessity for interdisciplinary research to assess the cleaning efficacy, economic implications, and environmental consequences of alternative emission reduction technologies, given the ambiguous environmental advantages of scrubbers [11]. There exists a significant performance disparity between theoretical modeling and actual in-situ implementation that necessitates resolution to validate results across various ship typologies [28]. Moreover, current research indicates that wet scrubbers possess limited efficacy in mitigating the health and environmental effects of ship emitted particulates, while the transference of pollutants like polycyclic aromatic hydrocarbons into marine ecosystems through effluent discharge demands a thorough assessment of both air and sea interfaces [17, 29]. Presently, there is a lack of comprehensive, measurement-informed life cycle assessments that evaluate the potential environmental impacts of heavy fuel oil and scrubber production and operation in comparison to low-sulfur fuels [30]. Another major technical problem is that artifact nanoparticles form in high humidity and sulfur environments. This needs new engineering solutions that work well for both SO_x and particulate abatement [17]. As a result, future studies should focus on creating integrated assessment frameworks that measure atmospheric dispersion, marine water quality degradation, and human health outcomes at the same time in order to find out how cost effective shower and dip wet scrubber technologies really are [29].

Despite this body of work, single-mode wet scrubbers exhibit intrinsic limits for multi-pollutant treatment. Immersion (dipping or bubble-column) scrubbers provide long gas residence time but a comparatively small interfacial area that is fixed by bubble size, whereas spray (shower) scrubbers create a very large droplet surface area but suffer from short gas–liquid contact time and droplet entrainment at high gas flow rates. Each mechanism therefore favors a different subset of pollutants and operating conditions, and neither alone performs consistently across the variable engine speeds typical of marine diesel operation.

Moreover, although combined spray-and-bubble concepts have been demonstrated at laboratory and pilot scale for flue gas, experimental data on a structurally combined "dipping + shower" contact pattern applied to small marine diesel (or simulated marine diesel) exhaust under varying engine speeds are still lacking. It remains unclear how much additional removal a combined contact pattern actually delivers for each gas (SO_2, CO, CO_2, NO_2), and how stable that performance is across the engine operating range.

This study addresses that gap through four specific contributions: (i) the design and construction of a low-cost prototype hybrid wet scrubber, fabricated largely from recycled industrial oil-storage drums, that integrates a dipping (bubble-column immersion) subsystem and a shower (spray atomization) subsystem in one unit; (ii) a systematic comparison of three operating modes (dipping-only, shower-only, and hybrid); (iii) simultaneous multi-pollutant measurement of SO_2, CO, CO_2, and NO_2 at the scrubber inlet and outlet; and (iv) performance testing at four engine-speed settings (25%, 50%, 75%, and 100% of rated speed). It should be emphasized that the term "hybrid" in this paper refers to this structural combination of bubble-immersion and spray gas–liquid contact within a single scrubber; it does not refer to the open-loop/closed-loop switching systems that are conventionally called hybrid scrubbers in the marine sector.

2 Materials and Methods

The Research & Development (R&D) methodology is a methodical, iterative process used in education, technology, and industry to bridge theory and practice. It usually entails needs analysis, design, prototyping,

testing (validation), revision, and small scale implementation [31].

2.1 Prototype Design

Figure 2 is the schematic diagram of the hybrid wet scrubber prototype coupled to a Dongfeng diesel engine. Exhaust gas enters via a submerged dip pipe in the dipping zone, rises through the water column and demister, passes the overhead spray zone, and exits from the top outlet. Circled numbers represent valves 1–8; S1 and S2 correspond to the inlet and outlet sampling points, and the dashed blue line indicates the operating liquid level.

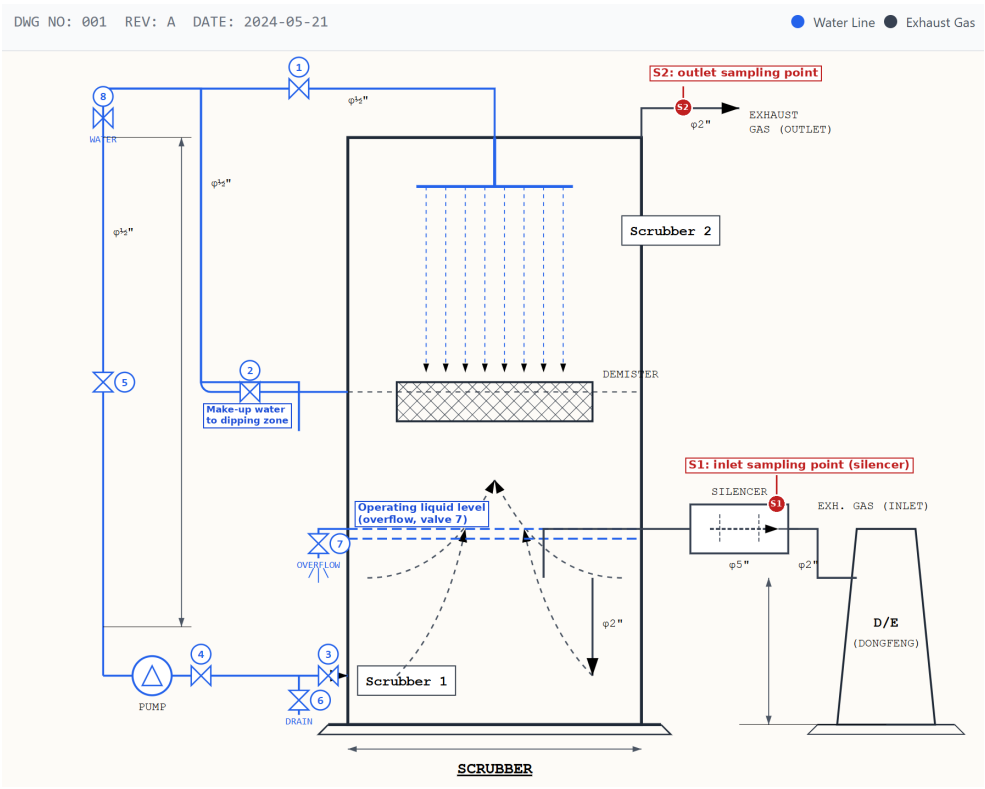


Figure 2. Schematic of the hybrid wet scrubber prototype coupled to a Dongfeng diesel engine

The wet scrubber is designed using many recycled materials from the industrial world, one of which is the use of used oil storage drums as the main body or tube of the wet scrubber system. This wet scrubber uses two gas washing systems: Scrubber system 1 uses the dipping wet scrubber method, where the exhaust silencer mounting is submerged in water, trapping the gas emissions in the water medium. Scrubber System 2 uses the shower wet scrubber method, where the gas coming out of the exhaust silencer is not immersed in water but is only sprayed with water coming out of the shower mounting that has been installed in the scrubber system, while Scrubber System 3 is a combination of scrubber systems 1 and scrubber systems 2.

Scrubber system 1 uses a depth in water system, which directs exhaust gases or pollutants into a volume of water. The exhaust gases are directed into the volume of water through a submerged Ø2 in inlet (dip) pipe that discharges near the vessel base and acts as a simple sparger; no Venturi throat is present. This process is also a method of trapping and absorbing exhaust pollutants through a static volume medium. The scrubber system 2 uses the Splashing Water system, which utilizes water splashes or sprays fed into the scrubber system filled with exhaust gas from Dongfeng diesel engines. These water splashes act as a medium for capturing and absorbing pollutant gases produced by diesel engines.

In all configurations, the exhaust gas follows the same path through the vessel (Figure 2): it leaves the silencer through a Ø2 in pipe that descends into the scrubber and discharges near the vessel base, rises through the water column (dipping zone), passes the demister pad, crosses the spray zone beneath the overhead shower header, and leaves through the Ø2 in outlet at the top. The three configurations were obtained with the valves shown in Figure 2: dipping-only, shower supply valve 1 closed and the vessel filled to the overflow level set by valve 7; shower-only, vessel drained through valve 6 below the outlet of the dip pipe so that the gas is not immersed, with valve 1 open; and hybrid, vessel filled to the overflow level with valve 1 open. Valve 2 supplies make-up water directly to the dipping zone when the liquid level needs topping up. The inlet sampling point (S1) was located at the silencer, upstream of

the scrubber, and the outlet sampling point (S2) at the Ø2 in scrubber outlet pipe; both points, the operating liquid level, and the make-up water line are marked in Figure 2.

2.2 Prototype Testing and Data Collection

In this paper, the treatment given to exhaust gas emissions from diesel engines was measured directly using the IoT-based multi-gas sensor node without a scrubber at varying engine-speed settings. The detailed experimental setup and operating conditions are summarized in Table 1. Furthermore, in the next experiment, exhaust gas emissions from the same diesel engine were passed through dipping and shower wet scrubber method, as well as a combination of both. In the first scenario, the engine system is turned on without using a scrubber, so that the measured exhaust emissions are the original emission values from the diesel engine exhaust through four different engine-speed settings, namely 25%, 50%, 75%, and 100%. In the second scenario, the engine system is turned on using scrubber system 1 so that the measured exhaust emissions are the emission values resulting from the scrubber system that has been created and acquired through four different engine-speed settings, namely 25%, 50%, 75%, and 100% of the rated speed of 2,200 rpm (i.e., 550, 1,100, 1,650, and 2,200 rpm). In the third scenario, the engine system is turned on using scrubber system 2 so that the measured exhaust emissions are the emission values resulting from the processing of the scrubber system that has been created and acquired through four different engine-speed settings, namely 25%, 50%, 75%, and 100% of rated speed. In the fourth scenario, the engine system is turned on using a combination of scrubbers 1 and 2 so that the measured exhaust emissions are the emission values resulting from the combined scrubber system that has been created and acquired through four different engine-speed settings.

Table 1. Experimental setup and operating conditions of the prototype hybrid wet scrubber

Parameter	Value	Parameter	Value
Engine model	Dongfeng S1115M, single-cylinder, horizontal four-stroke, direct injection	Scrubber body material	Recycled industrial oil-storage drum (steel), with demister pad between the immersion (dipping) zone and the overhead spray header (Figure 2)
Rated power	16.4 kW (22 hp) continuous; 17.9 kW (24 hp) maximum	Scrubber dimensions (D × H)	Ø360 mm × 660 mm (cylindrical body)
Rated speed	2,200 rpm	Water volume	≈ 34 L (filled to the overflow-port level, ≈ 330 mm above the base)
Test speed settings	25%, 50%, 75%, 100% of rated speed (550, 1,100, 1,650, 2,200 rpm)	Liquid level/immersion depth of dipping pipe	Level ≈ 330 mm (overflow-controlled); Ø50.8 mm (2 in) dipping pipe discharging near the vessel base
Fuel type	Automotive diesel fuel (Euro 5 standard)	Nozzle type and number	Single overhead shower-type spray head on a Ø12.7 mm ($\frac{1}{2}$ in) line
Fuel sulfur content	0.005% m/m (≈ 50 ppm)	Nozzle flow rate/spray pressure	Not measured (acknowledged limitation)
Exhaust flow rate	Not measured (acknowledged limitation)	Scrubbing liquid	Fresh water, without chemical additives
Exhaust temperature at inlet	≈ 90 °C (reduced to 30–40 °C at the scrubber outlet)	Duration of each test condition	60 min per operating condition

Gas concentrations at the scrubber inlet and outlet were measured with a purpose-built IoT-based multi-gas sensor node. The node consists of an ESP32 microcontroller with an I^2C multiplexer reading five gas sensors: a non-dispersive infrared (NDIR) MH-Z16 sensor for CO_2 (range 0–50,000 ppm, UART output, integrated temperature compensation) and factory-calibrated DFRobot electrochemical sensors for SO_2 (0–20 ppm), NO_2 (0–20 ppm), CO (0–1,000 ppm), and O_2 (single-point calibrated in ambient air at 20.9% vol). The electrochemical sensors apply built-in temperature-compensation algorithms, and the whole node was warmed up for at least five minutes (three minutes for the NDIR sensor) before each test. Sensor readings were streamed over MQTT (Mosquitto broker on a Raspberry Pi 5) and logged and visualized in Node-RED. The exhaust gas entered the scrubber at approximately 90 °C and was cooled to 30–40 °C at the outlet; the exhaust volumetric flow rate was not measured, which is acknowledged as a limitation when interpreting flow-related trends. At each engine-speed setting, the engine was first allowed to reach steady operation at the set speed before data acquisition, after which five consecutive readings were taken for each gas

at both sampling points from the data logged during the 60-min run. The time interval between these readings was not recorded in the dataset, which is acknowledged as a limitation. All efficiencies in this paper are reported as the mean of these five readings. Because consecutive readings from the same run may be autocorrelated, their standard deviation (SD) and coefficient of variation (CV) describe short-term measurement variability within a single run; they are not an estimate of the uncertainty that five independent experimental replicates would provide. It should be noted that this is a preliminary prototype test based on a single experimental campaign; independent replication of the whole test series was not performed, and the limited statistical support is acknowledged as a limitation of this study.

The electrochemical SO_2 and NO_2 sensors have a full-scale range of 20 ppm and the CO sensor of 1,000 ppm, and their output is capped at full scale. Of the 240 inlet readings, 34 were recorded at full scale (SO_2: 9, all at 100% of rated speed; NO_2: 22; CO: 3, all at 100% of rated speed), and 11 NO_2 outlet readings (all at 75% and 100% of rated speed) were also at full scale; no value above the range could be registered. These readings were retained as recorded, without correction or exclusion. Because a saturated inlet reading can only underestimate the true inlet concentration while the corresponding SO_2 and CO outlet readings were below full scale, the SO_2 and CO efficiencies at 100% of rated speed should be regarded as lower-bound estimates. For NO_2, both inlet and outlet readings were saturated in several cases, so the NO_2 efficiencies, particularly at 75% and 100% of rated speed, are not quantitatively reliable. Where a mean + SD error bar in Figures 3–5 extends beyond the full-scale value, this reflects the statistics of the recorded values, not a measured concentration above the sensor range.

2.3 Data Analysis

The analysis and evaluation process was conducted comprehensively through a direct measurement approach. The direct measurement approach was used to determine the extent to which scrubbers can reduce the percentage of exhaust pollutants from diesel engines. This performance parameter indicates the scrubber's ability to absorb gas pollutants produced by diesel engines. The more gas pollutants absorbed by the scrubber, the higher the efficiency of pollutant gas absorption. The efficiency percentage shows the comparison between the average emission value with a scrubber and the average emission value without a scrubber.

In this study, the concentrations of CO_2, SO_2, NO_2, and CO were recorded at both the inlet and the outlet of the scrubber, following the inlet–outlet comparison approach used in previous scrubber studies [30, 32]. Particulate matter, NO, and total SO_x were not measured. Outlet concentrations were observed during each run to confirm stable conditions before the readings were taken [18].

The reduction of each measured species (SO_2, NO_2, CO_2, and CO) was quantified by comparing its inlet and outlet concentrations. Because NO and total SO_x were not measured, the results refer to SO_2 and NO_2 only and do not represent total SO_x or NO_x removal. The removal efficiency was calculated as

$$\eta_i = \frac{C_{i,\text{inlet}} - C_{i,\text{outlet}}}{C_{i,\text{inlet}}} \times 100\% \tag{1}$$

where, $C_{i,inlet}$, and $C_{i,outlet}$ stand for the gas pollutant concentrations at the wet scrubber's inlets and outlets as measured by the IoT-based multi-gas sensor node [23].

In addition to the concentration of emission gases, the temperature of the diesel engine through the silencer, wet scrubber tube, and exhaust gas temperature from the wet scrubber were also collected to determine the contribution of temperature to the pollution produced.

3 Results and Discussion

The wet scrubber has been successfully constructed using two different methods integrated into one system. The first system is the dipping method, which works by feeding the engine exhaust gas directly through a silencer into the water already in the wet scrubber system. The second system is the splashing or shower method, which works by spraying water into the scrubber body filled with engine exhaust gas.

3.1 Dipping Method

This method uses a system that feeds exhaust gases (CO_2, SO_2, NO_2, and CO) from the engine directly into a volume of water. Comparison of exhaust gas concentration values based on engine-speed setting before and after wet scrubber filtration can be seen in Figure 3.

Wet gas scrubbing systems, also known as wet scrubbers, have long been regarded as a key technology for reducing industrial exhaust gas emissions. The dipping method offers a unique configuration in which the gas phase is dispersed into a continuous liquid phase [33]. The primary mechanism of this system is direct contact between the pollutant laden exhaust gas stream and the absorbent liquid, typically water or a specialized chemical solution. The absorption process allows contaminants to transfer from the gas phase to the liquid phase.

SO_2 has the highest efficiency and the best absorption efficiency among all gases, reaching an average of 78.7% at 25% of rated engine speed and decreasing to around 64–66% at higher engine speeds. With removal efficiency usually surpassing 95%, wet scrubbing is an essential technology for lowering SO_2 emissions in industrial and marine applications [23, 28]. In the dipping mode, exhaust gas is bubbled into the scrubbing liquid through a submerged inlet, as in a bubble column or jet bubbling reactor [34]. The immersion depth is the vertical distance between the gas outlet of this submerged pipe and the liquid surface; it is a geometric parameter that sets the bubble rise path and residence time and should be distinguished from the absorption mechanism itself. Compared to other exhaust gas components, SO_2 is highly soluble in water. When SO_2 exhaust gas comes into contact with the water in the scrubber, the SO_2 gas rapidly transfers from the gas phase to the liquid phase. Deeper immersion increases the residence time of gas bubbles in the liquid, allowing for more thorough absorption of SO_2 into the aqueous phase [34, 35]. Engine load and speed also have a big effect on how well lowering SO_2 exhaust gas concentration works. The more power the engine has to make, the more fuel it needs. As a result, the total amount of SO_2 released per hour goes up in a straight line as the load goes up. In general, raising the engine speed (particularly when the load stays the same or goes up) raises the fuel mass flow rate, which in turn raises the amount of SO_2 in the exhaust gas.

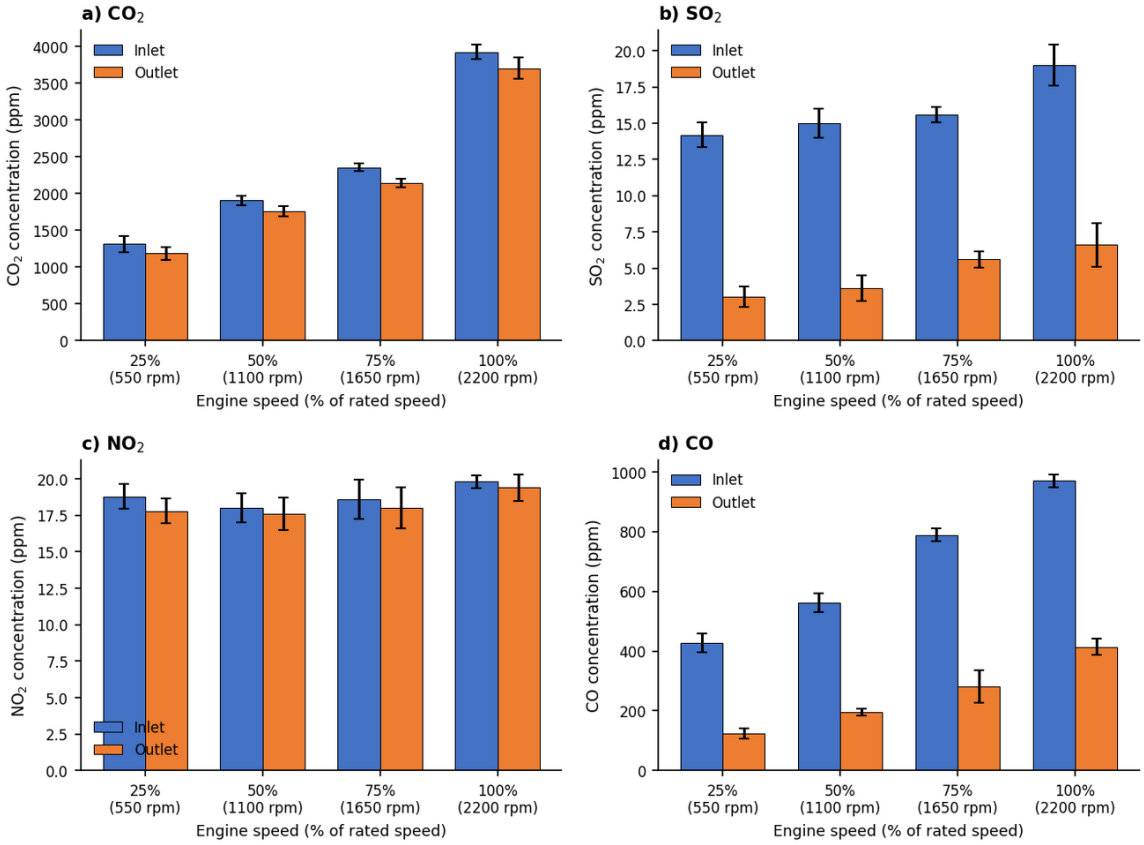


Figure 3. Inlet versus outlet exhaust gas concentrations (mean of five readings; error bars show ± 1 SD) for the dipping (bubble-immersion) method at four engine-speed settings (25–100% of rated speed): a) CO_2; b) SO_2; c) NO_2; and d) CO

CO removal is relatively high and decreases monotonically with engine speed, from 71.1% at 25% of rated speed to 57.3% at 100% of rated speed (Table 2). Higher engine speeds typically result in increased exhaust gas flow rates and temperatures, which can reduce the residence time of the gas within the scrubber and potentially lower the CO capture efficiency unless the solvent concentration or contact area is adjusted accordingly. This downward trend is reflected in the narrowing inlet–outlet gap in the grouped bar chart of Figure 3d and in Table 2: the higher the engine speed, the lower the efficiency. The inlet CO concentration (about 390–1,000 ppm) is the second highest of the measured gases after CO_2 (about 1,200–4,000 ppm). Compared to other pollutants such as SO_2 or ammonia (NH_3), carbon monoxide (CO) is less soluble in pure water. Therefore, the efficiency of CO removal in a bubble scrubber depends heavily on the interfacial contact area and an optimal mass transfer coefficient [36]. The CO removal efficiency is relatively high, ranging from 57.3% to 71.1%; this is because reduction efficiency is strongly influenced by physical parameters such as bubble diameter (which is determined by the sparger design), liquid column height, and gas surface velocity. Controlling bubble size through surface tension stabilization and preventing

coalescence is key to maximizing the mass transfer surface area. Although pure water alone is not sufficient to reduce CO, the incorporation of advanced oxidation systems (such as the Fenton process) or the use of copper-based absorbents (CuCl) in balloon column reactors can significantly improve CO reduction efficiency, often exceeding 80% in practical applications such as waste incinerators [37].

Table 2. Efficiency η of gas reduction (SO_2, CO, CO_2, NO_2) for dipping method (this study)

Gas	η Range (%)	Most Effective Speed	Consistency (CV Criterion)
SO_2	64.2–78.7	25%	Good (CV 3.6–8.8%)
CO	57.3–71.1	25%	Good (CV 4.1–9.5%)
CO_2	5.6–9.8	25–75%	Poor (CV 10.7–26.5%)
NO_2	2.1–5.3	25%	Poor (CV up to 137%)

Note: Efficiency ranges are the minimum and maximum of mean η across the four speed settings; consistency is classified from the coefficient of variation (CV) of the five consecutive readings at each speed setting, and each gas is assigned the category of its largest CV across the four settings: Good (CV < 10%), Moderate (CV 10–25%), Poor (CV > 25%).

CO_2 removal is low and inconsistent (mean 5.6–9.8%; Table 2) because CO_2 is difficult to absorb by conventional water-based wet scrubbers. The 25% speed setting provides the highest mean efficiency (9.8%), while the 100% setting provides the lowest (5.6%). Lower engine speeds, on the other hand, prolong the gas residence time, which may improve absorption efficiency. However, the decreased flow momentum may hinder the gas-liquid mixing intensity necessary for ideal mass transfer [38]. The temperature and flow rate of exhaust gas are changed by engine speed, which affects the scrubber's residence time [18]. Greater engine speeds enhance flow momentum but shorten the time bubbles have to react with the solvent, which could affect efficiency if the system isn't tuned for these higher loads [38]. Standard wet scrubbers typically have very little effect on removing CO_2 from ship exhaust. This is mostly because CO_2 is poorly soluble in the aqueous solutions, like seawater, that are commonly used in these scrubbers. Less than 10% CO_2 absorption efficiency has been demonstrated by wet scrubbing methods using specific urea solutions. Conventional wet scrubbers (which use water or weak alkaline solutions such as NaOH) have difficulty efficiently removing CO_2 due to its low solubility in water. Unlike SO_2, which is highly acidic and easily absorbed by alkaline solutions, CO_2 is a weak acid and does not readily dissolve or react with water or weak alkaline absorbents.

Figure 3c shows that the inlet–outlet difference for NO_2 is relatively small. Very low and variable efficiency NO_2 shows the lowest efficiency (2–5%), with many values of 0% (meaning no absorption at all in some measurements). This phenomenon occurs due to the low solubility of NO_2 in water. Although NO_2 is about five times more soluble in the aqueous phase than NO, its solubility is still insufficient to allow for significant absorption and reduction of NO_x from exhaust gases. Furthermore, NO is the primary component of NO_x and has low solubility in aqueous solutions, making it difficult to remove from exhaust gases through scrubbing alone [39]. Since NO can only be partially reduced under atmospheric conditions, pure water without chemical agents can absorb NO_2 with very little efficiency. Even after NO is oxidized to NO_2, plain water absorbs NO_2 only slowly, so a subsequent reduction stage with agents such as sulfite or thiosulfate is typically required [40]. The present NO_2 results are consistent with literature showing that pure-water wet scrubbing has only a limited capability for NO_x control, particularly when NO dominates the NO_x fraction; meaningful NO_x removal generally requires oxidants, alkaline absorbents, or other supporting methods. Because NO was not measured in this study, total NO_x removal could not be determined. Reported approaches for enhancing NO_x removal include reducing agents [41], an oxidation stage followed by sulfite or thiosulfate reduction [40], $NaClO_2$-based oxidative absorption [42], ozone pre-oxidation combined with wet absorption [43], and non-thermal plasma with catalytic oxidation and a wet electrostatic precipitator [44]; these options are compared in recent reviews [23, 45]. Their reported efficiencies depend strongly on absorbent concentration, pH, and gas composition and are therefore not directly comparable with the plain-water results of the present prototype.

3.2 Shower Method

The wet scrubber shower method uses spray nozzles that break the liquid into millions of fine droplets (atomization). This creates a very large surface area of contact between the exhaust gas and the liquid. The inlet and outlet exhaust gas concentrations under this shower method at different engine speeds are shown in Figure 4. The smaller the size of the water droplets, the greater the surface area to volume ratio, which significantly increases the rate of exhaust gas absorption. The system must be designed efficiently to balance high removal rates with economical pressure drop, which is generally recommended to be between 15 and 50 mmH_2O/m for the packing material to avoid a decline in machine performance [46, 47].

Optimal CO_2 removal efficiency is achieved at 75% of rated engine speed. Increasing the speed to 100% actually reduces efficiency significantly, likely due to excessive droplet atomization, which reduces the gas liquid contact time. The shower method in a wet scrubber works by dispersing the cleaning liquid through a nozzle to create fine droplets that fall through the gas stream. The CO_2 reduction efficiency of the wet scrubber system developed was only able to reduce the CO_2 emission concentration produced by the diesel engine by 10%. This is because the shower method tends to have moderate gas absorption efficiency compared to the bubble column or dipping methods [48]. Smaller particles increase the surface to volume ratio, thereby enhancing absorption capacity. However, if the particles are too small, the momentum of the gas flow can carry them out of the system, potentially reducing efficiency [49].

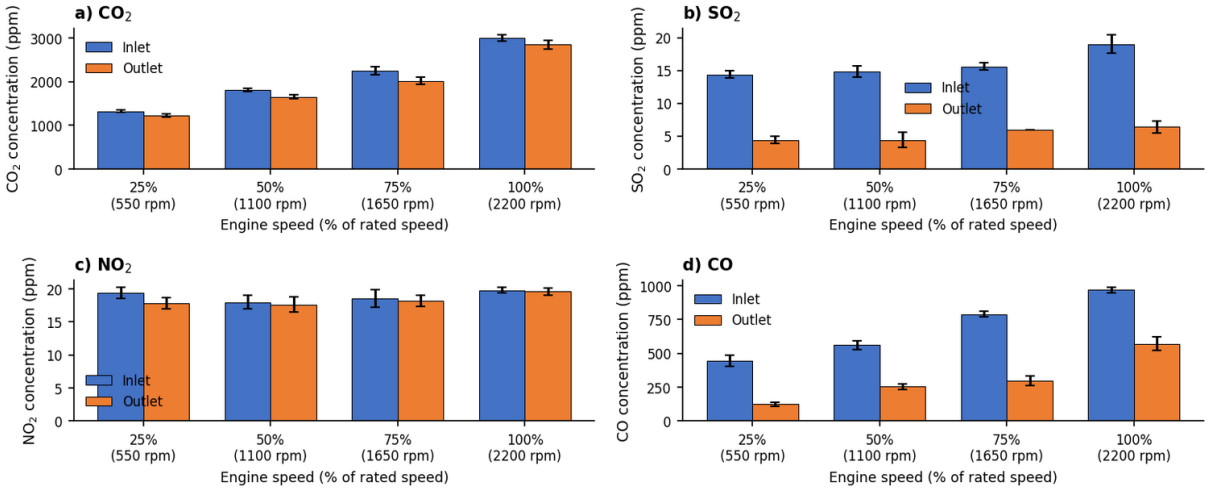


Figure 4. Inlet versus outlet exhaust gas concentrations (mean of five readings; error bars show ± 1 SD) for the shower (spray) method at four engine-speed settings (25–100% of rated speed): a) CO_2; b) SO_2; c) NO_2; and d) CO

SO_2 showed the highest mean efficiency at 50% of rated engine speed, but with high variability at that setting, indicating less consistent results. Overall, SO_2 has the highest efficiency of the measured gases, with mean values of 61.5–70.4% across the four speed settings (individual readings ranging from 60–79%, Table 3). This method works by spraying an absorbent liquid (usually water or an alkaline mixture such as lime) through a nozzle into the exhaust gas stream. When water droplets (spray) come into contact with the gas, mass transfer occurs, causing SO_2 to dissolve into the liquid droplets. High efficiency is achieved due to the small droplet size of the water stream; the smaller the water droplet size, the greater the total contact surface area between the liquid and the gas, thereby maximizing SO_2 absorption. Additionally, the higher the volume of liquid sprayed relative to the volume of passing gas, the higher the absorption efficiency [50]. The efficiency of SO_2 removal can be worsened by increasing the exhaust gas flow rate, which is frequently brought on by greater engine speeds. Even though the scrubber entrance is where most of the scouring takes place, a large increase in flow rate can cut the residence period in half, which could leave some of the gas untreated if the L/G ratio is not changed [18]. Spray hydrodynamics, liquid flow rates, and the tower's physical dimensions all have a significant impact on the effectiveness of SO_2 removal. The accessible surface area for absorption is determined by the droplet's velocity and distribution.

Table 3. Efficiency η of gas reduction (SO_2, CO, CO_2, NO_2) for shower method (this study)

Gas	η Range (%)	Most Effective Speed	Consistency (CV Criterion)
SO_2	61.5–70.4	50%	Good (CV 2.2–9.6%)
CO	41.1–72.3	25%	Moderate (CV 5.7–13.2%)
CO_2	4.8–10.1	75%	Moderate (CV 11.9–22.8%)
NO_2	1.0–8.2	25%	Poor (CV up to 224%)

Note: Efficiency ranges are the minimum and maximum of mean η across the four speed settings; consistency is classified from the coefficient of variation (CV) of the five consecutive readings at each speed setting, and each gas is assigned the category of its largest CV across the four settings: Good (CV < 10%), Moderate (CV 10–25%), Poor (CV > 25%).

NO_2 is removed appreciably only at low engine speed (25% of rated speed). An increase in engine speed results in a decrease in efficiency and significant variability, indicating that NO_2 is less soluble in water and sensitive to shower flow turbulence. A shower based scrubber's effectiveness starts with the amount of NO_2 that enters the

system, which is mostly determined by engine speed. The combustion chamber has a higher filling efficiency and a higher oxygen concentration at lower engine speeds, as reported for a different diesel engine at about 1,200 rpm [51].

In that engine, increasing the speed toward 3,200 rpm, which is above the 2,200 rpm rated speed of the engine used here, reduced the available oxygen and reaction time and hence the raw NO_x output; the same qualitative mechanism may contribute to the speed dependence observed in the present tests, although engine-out NO_x formation was not characterized in this study. The shower method frequently needs chemical or physical support to accomplish significant NO_2 reduction. For instance, NO_x removal efficiency of roughly 66% have been attained by systems that use ultra-low frequency wave treatment of the water [52]. Electrolyzed water is another method that can achieve reduction efficiencies of 43.2% to 48.9% [7].

CO shows the highest efficiency at 25% of rated speed (72.3%), but the trend with engine speed is not monotonic: efficiency falls to 54.5% at 50%, rises to 62.5% at 75%, and drops to 41.1% at 100% of rated speed (Table 3). Overall, lower engine speeds favour CO removal, consistent with the longer gas–liquid contact time at lower exhaust flow, while the intermediate rise at 75% indicates that factors other than flow, such as spray–gas mixing and inlet concentration, also play a role. Spray towers offer great collection efficiency for coarse PM because their primary method of particle collection is impaction. For particles larger than 5 μm, spray towers often have clearance rates of up to 90%. Particles with a diameter of 3 to 5 μm have removal efficiency ranging from 60 to 80%. Removal efficiencies drop to less than 50% below 3 μm. Applications for spray towers include dust management in fertilizer factories, pigment activities, and PM emissions from grinding operations.

3.3 Hybrid Systems (Dipping and Shower Methods)

Combining the dipping/bubble method with water spraying can significantly improve cleaning efficiency compared to using just one method. The inlet–outlet gas concentrations and removal efficiencies for this hybrid scheme are presented in Figure 5 and Table 4 respectively. In combined spray-and-bubble scrubbers reported in the literature, much of the pollutant load is removed in the first contact zone, while the second zone provides additional capacity for the remaining gas [48]. The very high efficiencies reported for such systems were obtained for specific pollutants, absorbents, and operating conditions; they are not results of the present prototype and cannot be attributed to the contact geometry alone.

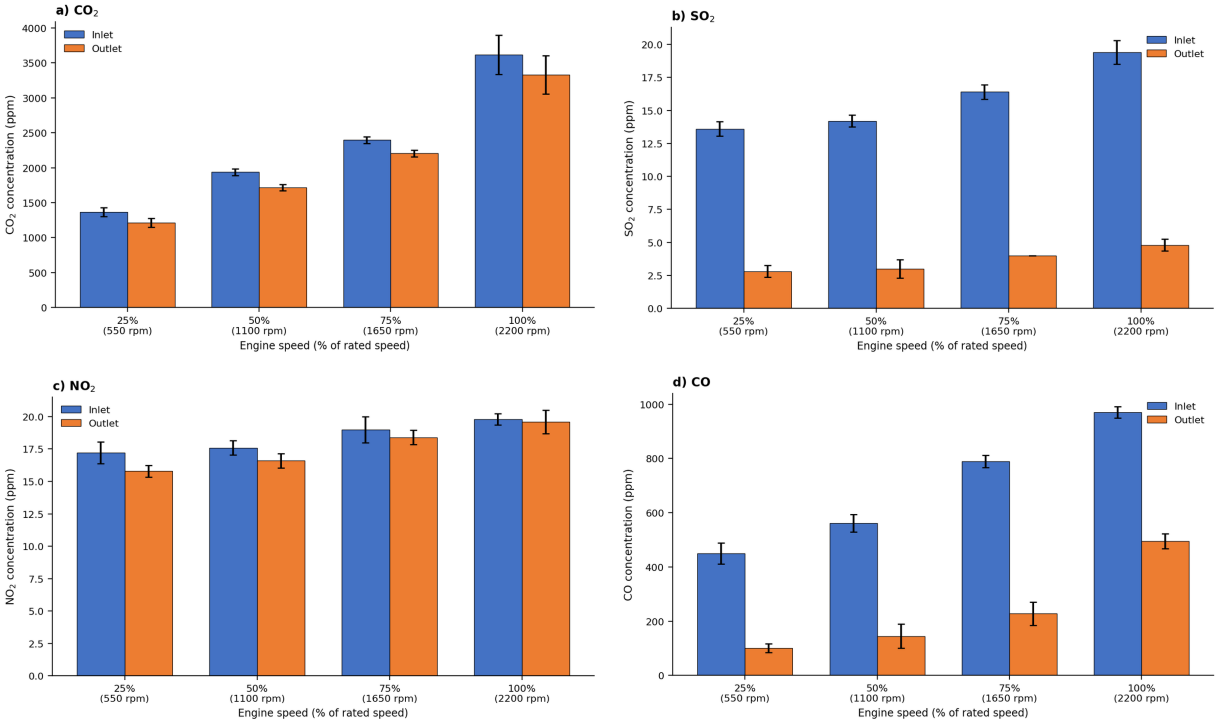


Figure 5. Inlet versus outlet exhaust gas concentrations (mean of five readings; error bars show ± 1 SD) for the hybrid method (simultaneous dipping + shower) at four engine-speed settings (25–100% of rated speed): a) CO_2; b) SO_2; c) NO_2; and d) CO

CO_2 efficiency remains relatively stable in the range of 8–11% across all engine-speed settings, with the differences between settings not particularly significant. SO_2 gas exhibits the highest efficiency, exceeding 75% at

all speed settings; notably, 25% of rated speed already provides very good efficiency (79.5%), indicating that the scrubber is highly effective for SO_2 even at low speeds. NO_2 efficiency is the lowest among the four gases, generally below 10%, and engine speed has almost no systematic effect on NO_2 reduction, so additional treatment methods are needed for this gas. CO efficiency is highest at 25% of rated speed (78%) and then decreases at higher speeds, most likely because of shorter gas–liquid contact time.

Table 4. Efficiency η of gas reduction (SO_2, CO, CO_2, NO_2) for hybrid method (this study)

Gas	η Range (%)	Most Effective Speed	Consistency (CV Criterion)
SO_2	75.2–79.5	25%	Good (CV 1.1–6.5%)
CO	49.0–77.8	25%	Moderate (CV 2.9–11.6%)
CO_2	8.0–11.3	25–50%	Moderate (CV 13.1–18.1%)
NO_2	1.1–8.1	25%	Poor (CV up to 224%)

Note: Efficiency ranges are the minimum and maximum of mean η across the four speed settings; consistency is classified from the coefficient of variation (CV) of the five consecutive readings at each speed setting, and each gas is assigned the category of its largest CV across the four settings: Good (CV < 10%), Moderate (CV 10–25%), Poor (CV > 25%).

The hybrid configuration that combines spray contact with a dipping/immersion zone can improve gas-liquid interaction because the spray section enlarges interfacial area through droplet formation, while the dipping section extends gas residence time and promotes secondary absorption. This explanation is consistent with a combined spray-and-bubble wet scrubbing concept, where simultaneous SO_2 and NO_x removal increased relative to single spray or single bubble operation because gas first undergoes removal in the spray zone and then further removal in the bubble/immersion zone. In that research, the combined system improved NO_x removal by up to 17–18% and SO_2 removal by 11–13% compared with separate units, indicating that hybrid contact patterns can strengthen pollutant capture [34]. Two differences limit the transferability of these gains. First, the flow path in the present prototype is reversed: the exhaust is first bubbled through the submerged inlet and only then passes the demister and the spray zone (Figure 2). Second, the cited system used ozone pre-oxidation, so its performance gain cannot be attributed to the contact geometry alone or transferred directly to plain-water operation.

The removal efficiency SO_2 is usually the highest among the tested gases because SO_2 is an acid gas that reacts quickly in alkaline liquid medium and has a relatively high solubility, it typically has the best removal efficiency of all the gases examined. Since SO_2 can be transformed into sulfite and bisulfite species after dissolving, absorption is regulated by both liquid-phase reactivity and physical dissolution. For this reason, wet scrubbers particularly spray towers are universally acknowledged as being highly successful in controlling SO_2. According to research on spray towers, the liquid-to-gas ratio, alkalinity/pH, inlet SO_2 concentration, and gas temperature all have a significant impact on SO_2 removal. According to a reviewed spray-tower study referenced in the literature, SO_2 removal might achieve 100% at inlet values of 500–1500 ppmv at an L/G ratio of roughly 3 L/m^3 and pH 11.7. Thus, the hybrid dipping–spray scrubber exhibiting the highest η for SO_2 is consistent with earlier research on wet scrubbers. Nevertheless, the hybrid efficiency of 75–79% obtained here remains clearly below the 95–98% values reported for full-scale marine and industrial scrubbers. We hypothesize that four prototype-specific factors contribute to this gap, although they were not tested individually in this study: (i) the scrubbing liquid was plain fresh water with no deliberate alkalinity, so absorption relied on physical dissolution rather than acid–base reaction; (ii) the compact drum geometry limits gas residence time compared with tall spray towers; (iii) the liquid-to-gas (L/G) ratio delivered by the single shower circuit, which could not be determined because neither the liquid nor the exhaust-gas flow rate was measured, may be lower than in high-efficiency spray towers; and (iv) the raw, uncooled exhaust enters the scrubber at high temperature, which reduces gas solubility. If confirmed, each of these factors could be addressed—through alkaline dosing, increased liquid level and spray flow, or pre-cooling—which would offer a route toward higher SO_2 capture in future versions of the device [53].

Although it can still be decreased in a wet scrubber under the right chemical conditions, NO_2 performance is typically lower than SO_2. The primary cause is that wet scrubbing nitrogen oxides is more difficult than wet scrubbing SO_2. Since NO has a low water solubility and dominates NO_x in many exhaust streams, efficient removal frequently necessitates previous oxidation to NO_2 or other more soluble forms. Oxidation potential and pH have a significant impact on the chemistry even in the presence of NO_2. A comparison of wet scrubbing for SO_2 and NO_x revealed that while effective NO_x capture required an oxidizing absorbent and an ideal pH of about 8 to balance oxidation and absorption, SO_2 removal was comparatively simple [23].

Because every pollutant absorbs differently in liquid media, the hybrid wet scrubber that combines dipping and spray techniques demonstrated varying gas reduction efficiencies. Wet scrubbing is one of the most efficient methods for removing sulfur because SO_2 is extremely soluble and reacts quickly in alkaline solution. Higher liquid to gas

ratios and alkaline circumstances can result in almost total SO_2 removal, according to spray tower research [53]. Although hybrid gas-liquid contact can still enhance performance by extending residence time and interfacial area, NO_2 removal is more challenging because nitrogen oxide absorption is highly dependent on oxidation-reaction routes and pH equilibrium. Because physical dissolution alone is weak, CO_2 reduction in wet scrubbers is typically limited unless the absorbent is strongly alkaline; literature reports that meaningful CO_2 capture requires high pH and reactive additives like $Ca(OH)_2$; CO, although removed comparatively well in this study thanks to the large interfacial area and extended residence time of the hybrid contact pattern, is intrinsically difficult to scrub because conventional wet scrubbers are not designed for CO capture; industrial CO separation typically relies on adsorption, membranes, cryogenic systems, or specialized chemical absorbents rather than water-based scrubbing, which explains why CO efficiency in this study fell sharply once exhaust flow increased [19]. Consequently, the basic physicochemical characteristics of the gases and earlier journal articles on wet scrubber systems are compatible with an efficiency trend observed here, namely $SO_2 > CO > CO_2 > NO_2$.

3.4 Comparative Performance of the Three Configurations

Table 5 summarizes the mean removal efficiency (η, $\pm$ SD from five replicate readings) of the four target gases under three scrubber configurations across four engine-speed conditions. The table also lists the percentage-point (pp) efficiency gain Δ of the hybrid configuration relative to the better-performing single mode; Δ is computed from the reported rounded mean values. Averaged over all four engine-speed settings, the hybrid mode improves SO_2 removal by about 6.1 percentage points over dipping-only and 10.4 percentage points over shower-only operation, and improves CO removal by about 3.5 and 10.4 percentage points, respectively. The gains for CO_2 (1.6–1.9 pp) and NO_2 (1.1–1.3 pp) are small in absolute terms, consistent with the solubility-limited behavior of these gases in pure water. The hybrid mode was, however, not superior at every individual condition: at 100% of rated speed its CO removal (49.0%) was lower than that of the dipping mode (57.3%), at 75% of rated speed its CO_2 removal (8.0%) was lower than that of the shower mode (10.1%), and for NO_2 the differences at 25%, 75%, and 100% of rated speed were negative but smaller than the SDs. Its advantage should therefore be read as a pollutant-specific average advantage rather than a consistent superiority.

Table 5. Mean removal efficiency (η) for three scrubber configurations at four engine speeds

Gas	Speed	Dipping η (% $\pm$ SD)	Shower η (% $\pm$ SD)	Hybrid η (% $\pm$ SD)	Δ Hybrid-Best Single (pp)
SO_2	25%	78.7 $\pm$ 5.5	69.4 $\pm$ 3.8	79.5 $\pm$ 3.0	+0.8
SO_2	50%	76.2 $\pm$ 4.5	70.4 $\pm$ 6.8	78.9 $\pm$ 5.1	+2.7
SO_2	75%	64.2 $\pm$ 2.3	61.5 $\pm$ 1.4	75.6 $\pm$ 0.8	+11.4
SO_2	100%	65.6 $\pm$ 5.7	66.4 $\pm$ 3.3	75.2 $\pm$ 2.4	+8.8
CO	25%	71.1 $\pm$ 4.3	72.3 $\pm$ 4.1	77.8 $\pm$ 2.3	+5.5
CO	50%	65.0 $\pm$ 3.5	54.5 $\pm$ 6.3	74.0 $\pm$ 8.6	+9.0
CO	75%	64.6 $\pm$ 6.1	62.5 $\pm$ 4.2	71.1 $\pm$ 6.0	+6.5
CO	100%	57.3 $\pm$ 2.4	41.1 $\pm$ 5.4	49.0 $\pm$ 2.8	-8.3
CO_2	25%	9.8 $\pm$ 1.0	7.4 $\pm$ 1.7	11.2 $\pm$ 1.8	+1.4
CO_2	50%	7.9 $\pm$ 1.3	8.7 $\pm$ 1.6	11.3 $\pm$ 2.0	+2.6
CO_2	75%	9.0 $\pm$ 1.0	10.1 $\pm$ 1.2	8.0 $\pm$ 1.1	-2.1
CO_2	100%	5.6 $\pm$ 1.5	4.8 $\pm$ 1.0	8.0 $\pm$ 1.1	+2.4
NO_2	25%	5.3 $\pm$ 0.2	8.2 $\pm$ 2.7	8.1 $\pm$ 2.8	-0.1
NO_2	50%	2.2 $\pm$ 3.1	2.2 $\pm$ 3.1	5.7 $\pm$ 0.2	+3.5
NO_2	75%	3.2 $\pm$ 3.0	2.0 $\pm$ 2.7	3.0 $\pm$ 2.8	-0.2
NO_2	100%	2.0 $\pm$ 2.8	1.0 $\pm$ 2.2	1.1 $\pm$ 2.4	-0.9

The hybrid mode is also the most stable configuration for SO_2: its CV across consecutive readings remains between 1.1% and 6.5% at every speed setting, and its mean efficiency varies only from 75.2% to 79.5% across the whole operating range, whereas the dipping mode falls from 78.7% to about 64% and the shower mode fluctuates between 61.5% and 70.4%. This indicates that combining spray contact (large interfacial area) with bubble immersion (long residence time) buffers the system against changes in exhaust flow.

The decline in CO removal at higher engine speeds supports the interpretation that the loss of efficiency is

flow-related. As engine speed rises from 25% to 100% of rated speed, the inlet CO concentration roughly doubles (about 430 to 970 ppm), reflecting the higher exhaust mass flow; over the same range, CO efficiency drops from 72.3% to 41.1% in shower-only mode but only from 77.8% to 49.0% in hybrid mode, and the immersion zone of the hybrid system partially compensates for the shorter droplet contact time. Because CO is only weakly soluble in water, its capture is especially sensitive to residence time, which explains why the flow effect is strongest for this gas.

3.5 Environmental Applicability and Limitations

From an environmental-management perspective, the relevance of this prototype lies in its accessibility. Because the device is built largely from recycled industrial oil-storage drums and operates with plain fresh water, it represents a low-capital, low-complexity auxiliary exhaust gas cleaning approach that could realistically be adapted for small vessels, port auxiliary equipment, harbor workboats, or stationary diesel generators in low-resource settings, where certified marine scrubbers or low-sulfur fuels may be economically out of reach. In such contexts, even partial abatement of SO_2 (about 75–79%) and CO (up to 78%) constitutes a meaningful local air-quality benefit.

The use of recycled oil drums also embodies a circular-use principle, extending the service life of industrial waste material as pollution-control equipment. However, this choice is appropriate only at the prototype and demonstration stage. Real shipboard exhaust gas cleaning systems cannot rely on non-standard containers: they must satisfy class and flag-state requirements for corrosion resistance against acidic condensate, pressure and vibration resistance, fire safety, maintainability, and back-pressure limits on the engine. Scaling this concept toward marine service would therefore require certified materials and a formal engineering assessment, while the recycled-drum approach remains valid for land-based or experimental applications.

Finally, wet scrubbing does not destroy pollutants; it transfers them from the gas phase to the liquid phase. Wash-water composition and pH were not measured in this study. Based on scrubbing chemistry and the literature on scrubber effluents, however, the recirculated water would be expected to accumulate dissolved SO_2 (as sulfurous acid, sulfite, and sulfate), dissolved CO_2, and nitrite/nitrate from absorbed NO_2, and therefore to become progressively more acidic. Combustion-derived particles may also be retained, but particulate collection is a separate mechanism from dissolved-gas absorption and was not quantified here. If discharged untreated, especially in ports or estuaries, such effluent could create secondary water pollution of the kind already documented for open-loop marine scrubber discharge [11, 15]. Responsible deployment of the proposed system requires closed-loop operation with periodic water replacement, neutralization (for example with lime or caustic dosing), sedimentation or filtration of particulates, and compliant disposal of the residue. Quantifying wash-water pH, sulfate, nitrate, and suspended-solid loads is identified as a priority for future work, so that the trade-off between air-pollution reduction and water-environment risk can be evaluated explicitly.

4 Conclusion

This study set out to determine whether a structural combination of bubble-immersion (dipping) and spray (shower) gas–liquid contact can outperform either contact mode alone for cleaning marine diesel exhaust. The prototype, built largely from recycled oil drums and tested at 25–100% of rated engine speed, answered this largely affirmatively: averaged over the four engine speeds, the hybrid configuration delivered the highest removal efficiencies of the three modes for all four gases and the most stable SO_2 removal, gaining on average about 6–10 percentage points for SO_2 and 3–10 percentage points for CO over the single-mode systems. The advantage was not universal, however: at 100% of rated speed the dipping mode removed more CO (57.3% vs 49.0%), and at 75% of rated speed the shower mode removed more CO_2 (10.1% vs 8.0%).

SO_2 and CO are the pollutants best suited to the current device. Hybrid-mode SO_2 removal remained at 75–79% across the entire speed range with a CV below 7%, and CO removal reached 78% at low engine speed, although it declined with increasing exhaust flow because of reduced gas–liquid contact time.

CO_2 and NO_2 remain the weak points of the system. With pure water as the scrubbing liquid, CO_2 removal was limited to 8–11% and NO_2 removal stayed below 10%, reflecting the low aqueous solubility of CO_2 and the dominance of poorly soluble NO in the NO_x fraction. The device should therefore be regarded as a transitional, auxiliary emission-control measure for SO_2 and partial CO abatement, not as a carbon-capture or NO_x-control technology; meeting those needs would require alkaline absorbents, oxidants, or a multi-stage treatment system.

The main limitations of this work are its single experimental campaign on one small engine, the use of fresh water only, the unquantified wash-water quality, and a prototype structure that does not yet meet marine equipment standards. Future work should test seawater and alkaline or oxidant-enhanced scrubbing solutions, add independently repeated experiments with full uncertainty analysis, measure wash-water pH and pollutant loads to manage the air-to-water pollution transfer, and evaluate long-term running stability, corrosion, and engine back-pressure before any practical deployment.

Author Contributions

Conceptualization, H.; investigation (data collection), H., A.R., and W.; resources (sensor preparation), A.R.; formal analysis, T.C. and W.; writing—original draft preparation, A.R.; supervision, T.C. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

References

[1] International Maritime Organization, "Resolution MEPC.304(72): Initial IMO strategy on reduction of GHG emissions from ships," 2018. https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MEPCDocuments/MEPC.304(72).pdf

[2] International Maritime Organization, "Resolution MEPC.377(80): 2023 IMO strategy on reduction of GHG emissions from ships," 2023. https://wwwcdn.imo.org/localresources/en/MediaCentre/PressBriefings/Documents/Resolution%20MEPC.377(80).pdf

[3] C. Zhang, J. Zhu, H. Guo, S. Xue, X. Wang, Z. Wang, T. Chen, L. Yang, X. Zeng, and P. Su, "Technical requirements for 2023 IMO GHG strategy," *Sustainability*, vol. 16, no. 7, p. 2766, 2024. https://doi.org/10.3390/su16072766

[4] International Council on Clean Transportation (ICCT), "Vision 2050: Fuel standards to align international shipping with the Paris agreement," 2025. https://theicct.org/publication/vision-2050-fuel-standards-to-align-international-shipping-with-the-paris-agreement-mar25/

[5] J. Chen, Y. Fei, and Z. Wan, "The relationship between the development of global maritime fleets and GHG emission from shipping," *J. Environ. Manage.*, vol. 242, pp. 31–39, 2019. https://doi.org/10.1016/j.jenvman.2019.03.136

[6] P. T. Aakko-Saksa, K. Lehtoranta, N. Kuittinen, A. Järvinen, J. Jalkanen, K. Johnson, H. Jung, L. Ntziachristos, S. Gagné, C. Takahashi *et al.*, "Reduction in greenhouse gas and other emissions from ship engines: Current trends and future options," *Prog. Energy Combust. Sci.*, vol. 94, p. 101055, 2023. https://doi.org/10.1016/j.pecs.2022.101055

[7] Y. Ryu, T. Kim, J. Kim, and J. Nam, "Investigation on the emission characteristics with a wet-type exhaust gas cleaning system for marine diesel engine application," *J. Mar. Sci. Eng.*, vol. 8, no. 11, p. 850, 2020. https://doi.org/10.3390/jmse8110850

[8] A. I. Kamal, R. B. Wicaksono, M. David, B. Hidayat, and S. Nurman, "Optimalisasi scrubber dalam menurunkan kadar sulfur dioksida di kapal MV. CMA CGM ELBE," *J. Marit. Malahayati*, vol. 5, no. 2, pp. 218–227, 2024. https://doi.org/10.70799/jumma.v5i2.100

[9] N. T. A. Othman and N. A. M. Alias, "Simulation analysis on SO_2 reduction inside a seawater scrubber," *J. Teknol.*, vol. 81, no. 6, 2019. https://doi.org/10.11113/jt.v81.13719

[10] L. F. E. D. Santos, K. Salo, and E. S. Thomson, "Quantification and physical analysis of nanoparticle emissions from a marine engine using different fuels and a laboratory wet scrubber," *Environ. Sci. Process. Impacts*, vol. 24, no. 10, pp. 1769–1781, 2022. https://doi.org/10.1039/D2EM00054G

[11] S. Endres, F. Maes, F. Hopkins, K. Houghton, E. M. Mårtensson, J. Oeffner, B. Quack, P. Singh, and D. Turner, "A new perspective at the ship-air-sea-interface: The environmental impacts of exhaust gas scrubber discharge," *Front. Mar. Sci.*, vol. 5, p. 139, 2018. https://doi.org/10.3389/fmars.2018.00139

[12] T. Zis, P. Angeloudis, M. G. H. Bell, and H. N. Psaraftis, "Payback period for emissions abatement alternatives: Role of regulation and fuel prices," *Transp. Res. Rec.*, vol. 2549, no. 1, pp. 37–44, 2016. https://doi.org/10.3141/2549-05

[13] D. Topali and H. N. Psaraftis, "The enforcement of the global sulfur cap in maritime transport," *Marit. Bus. Rev.*, vol. 4, no. 2, pp. 199–216, 2019. https://doi.org/10.1108/MABR-12-2018-0050

[14] M. E. L. Lambas, S. Ricci, and L. Rizzetto, "Reduction of gaseous emissions from ships: Comparative overview of methods and technologies," *WIT Trans. Built Environ*, pp. 93–104, 2021. https://doi.org/10.2495/UT210081

[15] J. Teuchies, T. J. S. Cox, K. Van Itterbeeck, F. J. R. Meysman, and R. Blust, "The impact of scrubber discharge on the water quality in estuaries and ports," *Environ. Sci. Eur.*, vol. 32, no. 1, p. 103, 2020. https://doi.org/10.1186/s12302-020-00380-z

[16] A. Bursuc, C. Munteanu, and S. Rus, "Pollution reduction methods for an emergency diesel generator fitted onboard a large container vessel," *Land Forces Acad. Rev.*, vol. 29, no. 1, pp. 1–13, 2024. https://doi.org/10.2478/raft-2024-0009

[17] A. Grigoriadis, N. Kousias, A. Raptopoulos-Chatzistefanou, H. Salberg, J. Moldanová, A. Hermansson, Y. Cha, A. Kontses, Z. Toumasatos, S. Mamarikas, and L. Ntziachristos, "Particulate and gaseous emissions from a large two-stroke slow-speed marine engine equipped with open-loop scrubber under real sailing conditions," *Atmosphere*, vol. 15, no. 7, p. 845, 2024. https://doi.org/10.3390/atmos15070845

[18] D. Flagiello, M. Esposito, F. Di Natale, and K. Salo, "A novel approach to reduce the environmental footprint of maritime shipping," *J. Mar. Sci. Appl.*, vol. 20, no. 2, pp. 229–247, 2021. https://doi.org/10.1007/s11804-021-00213-2

[19] P. Lu, X. Yan, L. Ye, D. Chen, D. Chen, J. Huang, and C. Cen, "Performance and mechanism of CO_2 absorption during the simultaneous removal of SO_2 and NO by wet scrubbing process," *J. Environ. Sci.*, vol. 135, pp. 534–545, 2024. https://doi.org/10.1016/j.jes.2022.08.028

[20] M. Tadros, M. Ventura, and C. G. Soares, "Optimization of the performance of marine diesel engines to minimize the formation of SO_x emissions," *J. Mar. Sci. Appl.*, vol. 19, no. 3, pp. 473–484, 2020. https://doi.org/10.1007/s11804-020-00156-0

[21] American Bureau of Shipping (ABS), "Advisory on exhaust gas scrubber systems," 2018. https://ww2.eagle.org/content/dam/eagle/advisories-and-debriefs/exhaust-gas-scrubber-systems-advisory.pdf

[22] R. Bergqvist, M. Turesson, and A. Weddmark, "Sulphur emission control areas and transport strategies—The case of Sweden and the forest industry," *Eur. Transp. Res. Rev.*, vol. 7, no. 2, p. 10, 2015. https://doi.org/10.1007/s12544-015-0161-9

[23] T. Chin, I. C. Tam, and C. Y. Yin, "Comparison of various chemical compounds for the removal of SO_2 and NO_x with wet scrubbing for marine diesel engines," *Environ. Sci. Pollut. Res.*, vol. 29, no. 6, pp. 8873–8891, 2022. https://doi.org/10.1007/s11356-021-16155-9

[24] N. Hasanspahic, S. Vujičić, L. Campara, and K. Piekarska, "Sustainability and environmental challenges of modern shipping industry," *J. Appl. Eng. Sci.*, vol. 19, no. 2, pp. 369–374, 2021. https://doi.org/10.5937/jaes0-28681

[25] C. C. Liao, J. L. Wu, Y. F. Wang, K. L. Huang, S. W. Huang, and Y. M. Kuo, "Advanced scrubbing/filtration devices for pollution control in medical laboratory waste incineration system," *Aerosol Air Qual. Res.*, vol. 24, no. 4, p. 230192, 2024. https://doi.org/10.4209/aaqr.230192

[26] B. Kareem Naser, M. Dauwed, A. Alkhayyat, M. Musa Jaber, S. Alyousif, and M. Hasan Ali, "An efficient method for heat recovery process and temperature optimization," *Comput. Mater. Contin.*, vol. 75, no. 1, pp. 1017–1031, 2023. https://doi.org/10.32604/cmc.2023.033957

[27] C. Novickis, D. J. O. Ferreira, S. W. Park, and M. C. Codolo, "Study of SO_2 absorption in a spray tower using CFD," *Research Square Preprint*, 2022. https://doi.org/10.21203/rs.3.rs-2327144/v1

[28] A. Romano and Z. Yang, "Decarbonisation of shipping: A state of the art survey for 2000–2020," *Ocean Coast. Manag.*, vol. 214, p. 105936, 2021. https://doi.org/10.1016/j.ocecoaman.2021.105936

[29] L. Anders, M. Bauer, S. Jeong, M. Schmidt, H. Hakkim, A. Kalamaščnikovs, E. I. Rosewig, J. Schade, R. Irsig, S. Ehlert *et al.*, "Limited efficiency of wet scrubbers in reducing the environmental impact of ship-emitted particles," *Environ. Chem. Lett.*, vol. 23, no. 4, pp. 923–929, 2025. https://doi.org/10.1007/s10311-025-01830-x

[30] P. M. Stathatou, I. Petrunia, T. Barenthin, G. Gotsis, P. Jeffrey, C. Fee, S. Bergeron, M. Tsezos, M. Triantafyllou, and N. Gershenfeld, "Marine scrubbers vs low-sulfur fuels: A comprehensive well-to-wake life cycle assessment supported by measurements aboard an ocean-going vessel," *Environ. Sci. Technol.*, vol. 59, no. 14, pp. 7066–7080, 2025. https://doi.org/10.1021/acs.est.4c10006

[31] M. Benny Aprial, I. E. P. Sari, A. E. Lubis, B. Helmi, and R. Aditya, "Development of a start basic engineering manual (bunch start, medium start, long start) for athletic court students," *ACPES J. Phys. Educ. Sport Health*, vol. 1, no. 1, pp. 31–40, 2021. https://doi.org/10.15294/ajpesh.v1i1.46298

[32] J. Bendl, M. R. Saraji-Bozorgzad, U. Käfer, S. Padoan, A. Mudan, U. Etzien, B. Giocastro, J. Schade, S. Jeong, E. Kuhn, M. Sklorz, C. Grimmer, T. Streibel, B. Buchholz, R. Zimmermann, and T. Adam, "How do different marine engine fuels and wet scrubbing affect gaseous air pollutants and ozone formation potential from ship emissions?" *Environ. Res.*, vol. 260, p. 119609, 2024. https://doi.org/10.1016/j.envres.2024.119609

[33] A. A. Abdulwahid, R. Situ, and R. J. Brown, "Underground diesel exhaust wet scrubbers: Current status and future prospects," *Energies*, vol. 11, no. 11, p. 3006, 2018. https://doi.org/10.3390/en11113006

[34] T. Si, C. Wang, X. Yan, Y. Zhang, Y. Ren, J. Hu, and E. J. Anthony, "Simultaneous removal of SO_2 and NO_x by a new combined spray-and-scattered-bubble technology based on preozonation: From lab scale to pilot scale," *Appl. Energy*, vol. 242, pp. 1528–1538, 2019. https://doi.org/10.1016/j.apenergy.2019.03.186

[35] W. Qirong, W. Mengli, D. Yungui, L. Cunfang, and C. Jianhong, "Enhanced efficiency of the sieve tray in a

desulfurization spray scrubber," *Front. Energy Res.*, vol. 10, p. 918233, 2022. https://doi.org/10.3389/fenrg.2022.918233

[36] Y. Wang and Y. Liu, "Oxidation removal of CO from flue gas using two Fenton-like wet scrubbing systems," *Energy Fuels*, vol. 33, no. 4, pp. 2961–2966, 2019. https://doi.org/10.1021/acs.energyfuels.8b04386

[37] W. Song, L. Zheng, K. Guo, H. Liu, W. Li, X. Guo, Y. Huang, and C. Liu, "Characteristics of bubbly flow, mass transfer, and CO_2 absorption in a two-stage internal loop airlift reactor assembly with a diamond-shaped guide baffle," *Ind. Eng. Chem. Res.*, vol. 63, no. 11, pp. 4952–4964, 2024. https://doi.org/10.1021/acs.iecr.3c04139

[38] X. Luo, M. Wang, J. Lee, and J. Hendry, "Dynamic modelling based on surface renewal theory, model validation and process analysis of rotating packed bed absorber for carbon capture," *Appl. Energy*, vol. 301, p. 117462, 2021. https://doi.org/10.1016/j.apenergy.2021.117462

[39] Z. Han, S. Yang, D. Zheng, X. Pan, and Z. Yan, "An investigation on NO removal by wet scrubbing using $NaClO_2$ seawater solution," *SpringerPlus*, vol. 5, no. 1, p. 751, 2016. https://doi.org/10.1186/s40064-016-2528-3

[40] T. Chin, I. Tam, and C. Y. Yin, "Wet scrubbing process with oxidation and reduction in series for removal of SO_2 and NO from marine diesel engine exhaust," *Chem. Eng. J.*, vol. 464, p. 142299, 2023. https://doi.org/10.1016/j.cej.2023.142299

[41] Y. Lee, J. H. Sung, B. Han, Y. J. Kim, and H. J. Kim, "Minimizing the consumption of reducing agents for NO_x removal in a wet scrubber without H_2S formation," *Sep. Purif. Technol.*, vol. 282, p. 120101, 2022. https://doi.org/10.1016/j.seppur.2021.120101

[42] J. Wei, Y. Luo, P. Yu, B. Cai, and H. Tan, "Removal of NO from flue gas by wet scrubbing with $NaClO_2$/$(NH_2)_2CO$ solutions," *J. Ind. Eng. Chem.*, vol. 15, no. 1, pp. 16–22, 2009. https://doi.org/10.1016/j.jiec.2008.07.010

[43] Z. Xu, X. Xiao, Y. Jia, P. Fang, J. Huang, H. Wu, Z. Tang, and D. Chen, "Simultaneous removal of SO_2 and NO by O_3 oxidation combined with wet absorption," *ACS Omega*, vol. 5, no. 11, pp. 5844–5853, 2020. https://doi.org/10.1021/acsomega.9b04031

[44] S. Cui, Z. Zhong, Y. Liao, L. Qi, and D. Fu, "Simultaneous removal of NO and SO_2 via an integrated system of nonthermal plasma combined with catalytic oxidation and wet electrostatic precipitator," *Energy Fuels*, vol. 33, no. 10, pp. 10 078–10 089, 2019. https://doi.org/10.1021/acs.energyfuels.9b02342

[45] M. Zhao, P. Xue, J. Liu, J. Liao, and J. Guo, "A review of removing SO_2 and NO_x by wet scrubbing," *Sustain. Energy Technol. Assess.*, vol. 47, p. 101451, 2021. https://doi.org/10.1016/j.seta.2021.101451

[46] M. Issa, H. Ibrahim, A. Ilinca, and M. Y. Hayyani, "A review and economic analysis of different emission reduction techniques for marine diesel engines," *Open J. Mar. Sci.*, vol. 9, no. 3, pp. 148–171, 2019. https://doi.org/10.4236/ojms.2019.93012

[47] A. Awoyomi, K. Patchigolla, and E. J. Anthony, "CO_2/SO_2 emission reduction in CO_2 shipping infrastructure," *Int. J. Greenh. Gas Control*, vol. 88, pp. 57–70, 2019. https://doi.org/10.1016/j.ijggc.2019.05.011

[48] B. R. Mohan and B. Meikap, "Performance characteristics of the particulate removal in a novel spray-cum-bubble column scrubber," *Chem. Eng. Res. Des.*, vol. 87, no. 1, pp. 109–118, 2009. https://doi.org/10.1016/j.cherd.2008.05.011

[49] A. Wansom, P. Maneechot, N. Jiteurtragool, and T. Vitidsant, "PM2.5 collection enhancement in a smart hybrid wet scrubber tower," *Processes*, vol. 11, no. 12, p. 3306, 2023. https://doi.org/10.3390/pr11123306

[50] O. Melnyk, S. Onyshchenko, O. Onishchenko, S. Kuznichenko, M. Bulgakov, and S. Kurdiuk, "Marine scrubber technologies as a transitional solution within the Earth system after IMO 2020," *Front. Earth Sci.*, vol. 14, p. 1784618, 2026. https://doi.org/10.3389/feart.2026.1784618

[51] M. Górska and T. Skrzek, "Nitrogen oxides emissions from a diesel engine fuelled with water-diesel microemulsions," *Adv. Sci. Technol. Res. J.*, vol. 14, no. 3, pp. 83–92, 2020. https://doi.org/10.12913/22998624/122716

[52] M. Issa, H. Ibrahim, R. Lepage, and A. Ilinca, "A review and comparison on recent optimization methodologies for diesel engines and diesel power generators," *J. Power Energy Eng.*, vol. 7, no. 6, pp. 31–56, 2019. https://doi.org/10.4236/jpee.2019.76003

[53] S. Darake, M. Hatamipour, A. Rahimi, and P. Hamzeloui, "SO_2 removal by seawater in a spray tower: Experimental study and mathematical modeling," *Chem. Eng. Res. Des.*, vol. 109, pp. 180–189, 2016. https://doi.org/10.1016/j.cherd.2015.11.027